DETERMINANTS

SUBJECT: MATHEMATICS **DATE: 09-04-2024**

Q.1 The set of equations $x - y + 3z = 2$, $2x - y + z = 4$, $x - 2y + \alpha z = 3$ has

(A) unique soluton only for $\alpha = 0$

(B) unique solution for $\alpha \neq 8$

(C) infinite number of solutions for $\alpha = 8$

(D) no solution for $\alpha = 8$

Q.2 A solution of the system $x + y - z = 0$, $x^2 + y^2 + z^2 = 0$, $x^4 + y^4 + z^4 = 0$ is

(A) $(1 , \omega , \omega^2)$

(B) $(-1, -\omega , \omega^2)$

(C) $(1 , \omega , -\omega^2)$

(D) the trivial solution, $\omega \neq 1$ being a cube root of unity

Q.3 The determinant $\begin{vmatrix} a^2 & a^2-(b-c)^2 & bc \\ b^2 & b^2-(c-a)^2 & ca \\ c^2 & c^2-(a-b)^2 & ab \end{vmatrix}$ is divisible by :

(A) $a+b+c$

(B) $(a+b)(b+c)(c+a)$

(C) $a^2+b^2+c^2$

(D) $(a-b)(b-c)(c-a)$

Q.4 The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \le A \le \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix} 1+\sin^2 A & \cos^2 A & 2\sin 4\theta \\ \sin^2 A & 1+\cos^2 A & 2\sin 4\theta \\ \sin^2 A & \cos^2 A & 1+2\sin 4\theta \end{vmatrix} = 0 \text{ are :}$$

(A) $A = \frac{\pi}{4}$, $\theta = -\frac{\pi}{8}$

(B) $A = \frac{3\pi}{8} = \theta$

(C) $A = \frac{\pi}{5}$, $\theta = -\frac{\pi}{8}$

(D) $A = \frac{\pi}{6}$, $\theta = \frac{3\pi}{8}$

Q.5 Given $p, q \in \{0, 1, 2, 3, 4, ..., 19, 20\}$. Consider the system of equations

$x + y + z = 4$

$2x + y + 3z = 6$

$x + 2y + pz = q$

Let **L:** denotes number of ordered pairs (p, q) so that the system of equations has unique solution,

M: denotes number of ordered pairs (p, q) so that the system of equations has no solution and

N: denotes number of ordered pairs (p, q) so that the system of equations has infinite solutions.

Find $(L + M - N)$.

Q.6 Let $f(x) = \begin{vmatrix} 1 & -1 \\ -x & x^2-2 \end{vmatrix} + \begin{vmatrix} -1 & 1 \\ -2x & 2x^2-4 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -3x & 3x^2-6 \end{vmatrix} + \begin{vmatrix} -1 & 1 \\ -4x & 4x^2-8 \end{vmatrix} + ...$

$..............+ \begin{vmatrix} 1 & -1 \\ -31x & 31x^2-62 \end{vmatrix}$

If P: represent minimum value of g(x) where $g(x) = \int_0^x f(x-t)\,dt$, $x \in [0, 3]$

and Q : represent the value of (a + b), if the set of values of k for which the equation $|f(x^2)| = k$ has six different solutions is (a, b) , then find the value of (Q – 3P).

Q.7 Let $D_1 = \begin{vmatrix} x & a & b \\ -1 & 0 & x \\ x & 2 & 1 \end{vmatrix}$ and $D_2 = \begin{vmatrix} cx^2 & 2a & -b \\ x & 2 & 1 \\ -1 & 0 & x \end{vmatrix}$.

If all the roots of the equation $(x^2 - 4x - 7)(x^2 - 2x - 3) = 0$ satisfies the equation $D_1 + D_2 = 0$ then find the value of $(a + 4b + c)$.

Q.8 If $p^2 + q^2 + r^2 = -2$ and $g(x) = \begin{vmatrix} 1+p^2x & (1+q^2)x & (1+r^2)x \\ (1+p^2)x & 1+q^2x & (1+r^2)x \\ (1+p^2)x & (1+q^2)x & 1+r^2x \end{vmatrix}$ for all $x \in R$ then find the value of

definite integral $\int_1^4 g(x)\,dx$.

ANSWER KEY												
Ques.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**				
Ans.	**BD**	**BCD**	**ACD**	**ABCD**	**439**	**228**	**0**	**9**				